

MILK RIVER RIDGE RESERVOIR, AB, Canada

WMO: 710150

Lat: 49.2839N

Lon: 112.5383W

Elev: 1305

StdP: 86.60

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.4	-24.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.0	N/A	27.1	N/A	25.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A		-30.1	32.1	3.8	2.4	-32.8	33.8	-35.0	35.2	-37.1	36.6	-39.9	38.3
				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-5.3	-7.4	-1.9	3.6	9.4	13.2	17.5	17.0	12.3	5.4	-0.2	-6.6		
	DBStd	10.60	7.95	8.74	7.11	5.44	4.70	3.66	3.52	3.86	5.39	5.58	5.99	7.86		
	HDD10.0	2634	474	488	369	201	69	11	2	36	161	307	516			
	HDD18.3	5007	732	721	628	443	276	157	57	70	192	402	555	774		
	CDD10.0	746	0	0	0	8	52	108	233	220	104	17	2	0		
	CDD18.3	76	0	0	0	0	1	4	32	30	10	0	0	0		
	CDH23.3	866	0	0	0	2	14	38	347	346	117	2	0	0		
	CDH26.7	211	0	0	0	0	0	4	91	92	24	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	379	12	10	18	31	57	90	40	36	32	22	16	14		
	PrecMax	579	21	22	40	62	106	225	139	83	93	69	34	29		
	PrecMin	204	2	1	4	13	16	15	2	1	3	5	4	1		
	PrecStd	89	5	5	9	12	25	51	31	21	23	15	8	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	10.2	13.5	22.0	25.1	27.1	31.8	31.4	29.4	22.2	16.1	9.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	7.4	7.0	10.6	17.3	22.8	24.1	29.1	29.1	26.7	18.2	11.7	6.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	5.5	5.0	8.5	14.3	20.4	22.5	27.0	27.2	24.0	15.4	9.5	4.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	4.1	3.4	6.1	11.7	17.9	20.4	25.0	25.1	21.1	12.9	7.4	2.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.6	8.8	10.3	10.5	10.4	12.1	12.6	11.4	9.6	8.5	7.6		
		MCDBR	8.5	9.6	11.4	14.1	13.9	12.8	14.5	15.6	15.3	12.9	11.0	8.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.249	0.272	0.305	0.317	0.316	0.318	0.334	0.300	0.270	0.247	0.227		
	taud		2.470	2.500	2.503	2.419	2.436	2.486	2.489	2.414	2.488	2.552	2.517	2.465		
	Ebn at Noon		854	930	955	946	942	943	935	903	908	889	840	825		
	Edh at Noon		58	72	86	105	108	104	102	105	87	68	55	50		
(44) All-Sky Solar Radiation	RadAvg		1.24	2.24	3.48	4.80	5.72	6.39	7.11	5.67	4.03	2.50	1.39	0.99		
	RadStd		0.10	0.17	0.30	0.32	0.50	0.49	0.39	0.36	0.33	0.24	0.11	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page